

Multiindex Notation and the Symbol of a PDE

MA 5565

Multiindex Notation

Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$.

- Order:

$$|\alpha| = \alpha_1 + \dots + \alpha_n$$

- Factorial:

$$\alpha! = \alpha_1! \dots \alpha_n!$$

- Monomial:

$$x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$$

- Partial derivative:

$$D^\alpha u = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}$$

Linear Differential Operators

Let $\Omega \subset \mathbb{R}^n$ be open.

A linear differential operator of order m has the form

$$Lu(x) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha u(x),$$

where $a_\alpha \in C^\infty(\Omega)$.

The **principal part** consists of the highest order terms:

$$L_m u = \sum_{|\alpha|=m} a_\alpha(x) D^\alpha u.$$

Definition of the Symbol

Let $\xi = (\xi_1, \dots, \xi_n) \in \mathbb{R}^n$.

Definition (Symbol).

The symbol of L is the polynomial

$$p(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha.$$

The **principal symbol** is

$$p_m(x, \xi) = \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha.$$

The principal symbol is homogeneous of degree m in ξ .

Interpretation of the Symbol

Heuristic substitution:

$$D_j \longleftrightarrow \xi_j$$

so that

$$D^\alpha \longleftrightarrow \xi^\alpha.$$

The principal symbol captures:

- propagation of singularities,
- characteristic directions,
- classification of equations.

Second Order Scalar PDE in Two Variables

Let $u = u(x, y)$.

A general second order linear PDE:

$$Lu = a(x, y)u_{xx} + 2b(x, y)u_{xy} + c(x, y)u_{yy} + \text{lower order terms.}$$

Principal symbol:

$$p_2(x, y, \xi) = a\xi_1^2 + 2b\xi_1\xi_2 + c\xi_2^2.$$

This is a quadratic form in $\xi = (\xi_1, \xi_2)$.

Matrix Form of the Principal Symbol

Write

$$p_2(x, y, \xi) = (\xi_1 \quad \xi_2) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}.$$

The classification depends on the discriminant

$$\Delta = b^2 - ac.$$

Elliptic Equations

Definition.

The equation is **elliptic** at (x, y) if

$$p_2(x, y, \xi) \neq 0 \quad \text{for all } \xi \neq 0.$$

For two variables this is equivalent to

$$b^2 - ac < 0.$$

Example:

$$u_{xx} + u_{yy} = 0 \quad (\text{Laplace equation})$$

Principal symbol: $\xi_1^2 + \xi_2^2$.

Hyperbolic Equations

Definition.

The equation is **hyperbolic** at (x, y) if

$$b^2 - ac > 0.$$

The quadratic form has two distinct real characteristic directions.

Example:

$$u_{xx} - u_{yy} = 0$$

Principal symbol:

$$\xi_1^2 - \xi_2^2.$$

Parabolic Equations

Definition.

The equation is **parabolic** at (x, y) if

$$b^2 - ac = 0.$$

The quadratic form has a repeated real root.

Example:

$$u_{xx} = 0 \quad (\text{degenerate example})$$

Time-dependent example:

$$u_t - u_{xx} = 0.$$

Geometric View

The classification corresponds to the geometry of

$$p_2(x, y, \xi) = 0.$$

- Elliptic: no real nonzero ξ solves $p_2 = 0$.
- Hyperbolic: two real characteristic directions.
- Parabolic: one repeated real direction.

The symbol determines characteristic curves and propagation behavior.

Summary

- Multiindex notation compactly encodes derivatives.
- The symbol replaces D^α by ξ^α .
- The principal symbol determines equation type.
- For second order scalar PDEs in two variables:

$$\Delta = b^2 - ac$$

classifies elliptic, hyperbolic, and parabolic behavior.

Characteristic Curves

For the second order equation

$$au_{xx} + 2bu_{xy} + cu_{yy} + \text{l.o.t.} = 0,$$

the principal symbol is

$$p_2(x, y, \xi) = a\xi_1^2 + 2b\xi_1\xi_2 + c\xi_2^2.$$

Characteristic directions are defined by

$$p_2(x, y, \xi) = 0, \quad \xi \neq 0.$$

If a curve $\phi(x, y) = \text{const}$ is characteristic, then its normal vector satisfies

$$\xi = \nabla\phi = (\phi_x, \phi_y).$$

Thus characteristic curves satisfy

Differential Equation for Characteristics

Assume $\phi_y \neq 0$ and set

$$\lambda = \frac{dx}{dy}.$$

Since $\nabla\phi$ is normal to the curve,

$$\frac{\phi_x}{\phi_y} = -\frac{dy}{dx}.$$

Substituting into the characteristic condition gives

$$a \left(\frac{dy}{dx} \right)^2 - 2b \left(\frac{dy}{dx} \right) + c = 0.$$

Equivalently,

$$a\lambda^2 - 2b\lambda + c = 0.$$

The number of real roots matches the sign of $b^2 - ac$

Canonical Form: Hyperbolic Case

If $b^2 - ac > 0$, there are two distinct real roots:

$$\lambda_1, \lambda_2.$$

Let $\xi(x, y)$ and $\eta(x, y)$ be functions constant along the two characteristic families.

Under the change of variables

$$(x, y) \mapsto (\xi, \eta),$$

the principal part becomes

$$u_{\xi\eta}.$$

Canonical hyperbolic form:

$$u_{\xi\eta} + \text{l.o.t.} = 0.$$

Canonical Form: Elliptic Case

If $b^2 - ac < 0$, there are no real characteristic directions.

One can locally choose new variables (ξ, η) so that the quadratic form diagonalizes with the same sign:

$$p_2 = \alpha(\xi, \eta)(\xi_1^2 + \xi_2^2).$$

Canonical elliptic form:

$$u_{\xi\xi} + u_{\eta\eta} + \text{l.o.t.} = 0.$$

Example: Laplace equation.

Canonical Form: Parabolic Case

If $b^2 - ac = 0$, there is one repeated real root.

There is a single characteristic family.

After a change of variables,

$$p_2 = \alpha(\xi, \eta) \xi_1^2.$$

Canonical parabolic form:

$$u_{\xi\xi} + \text{l.o.t.} = 0.$$

Example: Heat equation (after separating time variable).

Geometric Interpretation

The canonical reduction corresponds to diagonalizing the quadratic form

$$adx^2 + 2bdx\,dy + cdy^2.$$

- Elliptic: definite quadratic form.
- Hyperbolic: signature $(1, -1)$.
- Parabolic: degenerate.

Thus classification is determined entirely by the principal symbol.

The Cauchy Problem (Scalar Equation)

Let

$$Lu = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha u$$

be a linear differential operator of order m .

Let Σ be a smooth hypersurface given locally by

$$\phi(x) = 0, \quad \nabla \phi \neq 0.$$

Cauchy problem:

Find u such that

$$Lu = f \quad \text{near } \Sigma,$$

with prescribed initial data

$$D_\nu^k u|_\Sigma = g_k, \quad k = 0, \dots, m-1,$$

where ν is a normal direction to Σ .

Normal Derivatives

Let

$$\nu = \nabla \phi.$$

Define the normal derivative operator

$$D_\nu = \sum_{j=1}^n \nu_j \partial_{x_j}.$$

Initial data consist of

$$u|_\Sigma, \quad D_\nu u|_\Sigma, \quad \dots, \quad D_\nu^{m-1} u|_\Sigma.$$

Exactly m pieces of data are required for an order m equation.

Noncharacteristic Surfaces

Definition.

The hypersurface Σ is noncharacteristic at x_0 if

$$\rho_m(x_0, \nabla \phi(x_0)) \neq 0.$$

Equivalently, the highest-order normal derivative can be solved for in terms of tangential derivatives.

In this case the Cauchy problem is locally solvable (at least in analytic categories).

Higher Order Scalar Operators

Let

$$Lu = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha u$$

be a linear differential operator of order m .

Principal symbol:

$$p_m(x, \xi) = \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha.$$

This is a homogeneous polynomial of degree m in ξ .

Ellipticity: Equivalent Definitions

Definition. L is elliptic at x if

$$p_m(x, \xi) \neq 0 \quad \text{for all } \xi \neq 0.$$

Equivalent formulations:

- ① $p_m(x, \xi)$ has no nontrivial real zeros.
- ② There exists $C > 0$ such that

$$|p_m(x, \xi)| \geq C|\xi|^m \quad \text{for } |\xi| \text{ large.}$$

- ③ The highest-order part dominates all lower-order terms.

Ellipticity depends only on the principal symbol.

Examples of Higher Order Elliptic Operators

Biharmonic operator:

$$\Delta^2 u.$$

Principal symbol:

$$|\xi|^4.$$

Since $|\xi|^4 > 0$ for $\xi \neq 0$, the operator is elliptic.

More generally, any power Δ^k is elliptic.

Linear Systems

Let $u = (u_1, \dots, u_N)^T$.

A linear system of order m has the form

$$Lu = \sum_{|\alpha| \leq m} A_\alpha(x) D^\alpha u,$$

where each $A_\alpha(x)$ is an $N \times N$ matrix.

Matrix-Valued Symbol

The symbol is the matrix

$$P(x, \xi) = \sum_{|\alpha| \leq m} A_\alpha(x) \xi^\alpha.$$

The principal symbol:

$$P_m(x, \xi) = \sum_{|\alpha|=m} A_\alpha(x) \xi^\alpha.$$

This is an $N \times N$ matrix polynomial in ξ .

Ellipticity for Systems

Definition. The system is elliptic at x if

$$\det P_m(x, \xi) \neq 0 \quad \text{for all } \xi \neq 0.$$

Equivalently:

- $P_m(x, \xi)$ is invertible for all $\xi \neq 0$.
- There exists $C > 0$ such that

$$\|P_m(x, \xi)^{-1}\| \leq C|\xi|^{-m}.$$

Thus ellipticity reduces to invertibility of the principal symbol matrix.

Definition of Characteristic Surface

Let L be a scalar operator of order m .

A surface

$$\phi(x) = 0$$

is **characteristic at** x_0 if

$$p_m(x_0, \nabla \phi(x_0)) = 0, \quad \nabla \phi(x_0) \neq 0.$$

Otherwise the surface is **noncharacteristic**.

Interpretation

Heuristically:

Replace

$$D^\alpha \rightarrow \xi^\alpha.$$

If $\phi(x) = 0$ is characteristic, then the normal direction $\nabla\phi$ annihilates the principal symbol.

Characteristic surfaces are precisely those across which uniqueness of the Cauchy problem may fail.

Characteristic Polynomial

The characteristic equation is

$$p_m(x, \xi) = 0.$$

This determines characteristic directions.

Second order example (two variables):

$$a\phi_x^2 + 2b\phi_x\phi_y + c\phi_y^2 = 0.$$

These define the characteristic curves.

Characteristic Surfaces for Systems

For systems:

$$\det P_m(x, \xi) = 0$$

is the characteristic equation.

A surface $\phi(x) = 0$ is characteristic if

$$\det P_m(x, \nabla \phi(x)) = 0.$$

Thus characteristic directions are determined by the vanishing of the determinant of the principal symbol matrix.

Ellipticity and Characteristics

- Elliptic operators have no real characteristic directions.
- Hyperbolic operators possess real characteristic surfaces.
- Parabolic operators have degenerate characteristics.

Therefore, the geometry of

$$p_m(x, \xi) = 0$$

governs propagation and well-posedness.

Elliptic Equations and Time Evolution

Consider Laplace's equation:

$$\Delta u = 0.$$

Principal symbol:

$$p_2(\xi) = |\xi|^2.$$

There are *no real characteristic directions*.

If we attempt a time-evolution formulation:

$$u_{tt} + \Delta u = 0,$$

the symbol becomes

$$\tau^2 + |\xi|^2,$$

which never vanishes for real $(\tau, \xi) \neq 0$.

Consequences:

- No real characteristic cones.
- No finite-speed propagation.

Fourier Transform and the Symbol

Let

$$\widehat{u}(\xi) = \int_{\mathbb{R}^n} e^{-ix \cdot \xi} u(x) dx.$$

Key identity:

$$\widehat{D^\alpha u}(\xi) = (i\xi)^\alpha \widehat{u}(\xi).$$

Thus under Fourier transform,

$$Lu \longrightarrow p(x, \xi) \widehat{u}(\xi) \quad (\text{for constant coefficients}).$$

Interpretation:

- The symbol is the Fourier multiplier.
- Zeros of the principal symbol determine where inversion fails.
- Ellipticity means the multiplier never vanishes.

The symbol is therefore the algebraic shadow of differentiation in frequency space.

What the Symbol Tells Us

The principal symbol controls behavior at high frequency:

$$\widehat{D^\alpha u} = (i\xi)^\alpha \widehat{u}.$$

Key consequences:

- **Elliptic:** $|p(\xi)| \sim |\xi|^m$
 $\Rightarrow$ division improves decay (smoothing effect).
- **Hyperbolic:** real zeros of p
 $\Rightarrow$ oscillatory propagation along characteristic cones.
- **Parabolic:** partial degeneracy
 $\Rightarrow$ directional smoothing.

The symbol encodes how each frequency evolves.

Wave Equation in Two Spatial Dimensions

Consider

$$u_{tt} - c^2 \Delta u = 0, \quad x \in \mathbb{R}^2.$$

Initial data:

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x).$$

Principal symbol:

$$p_2(\tau, \xi) = \tau^2 - c^2 |\xi|^2.$$

Take Fourier transform in the spatial variables:

$$\widehat{u}(\xi, t) = \int_{\mathbb{R}^2} e^{-ix \cdot \xi} u(x, t) dx.$$

Reduction to an ODE in Frequency Space

Using

$$\widehat{\Delta} u = -|\xi|^2 \widehat{u},$$

the PDE becomes

$$\widehat{u}_{tt}(\xi, t) + c^2 |\xi|^2 \widehat{u}(\xi, t) = 0.$$

For each fixed ξ , this is a harmonic oscillator:

$$\widehat{u}_{tt} + \omega(\xi)^2 \widehat{u} = 0, \quad \omega(\xi) = c|\xi|.$$

Solution:

$$\widehat{u}(\xi, t) = \widehat{f}(\xi) \cos(c|\xi|t) + \frac{\widehat{g}(\xi)}{c|\xi|} \sin(c|\xi|t).$$

Plane Wave Interpretation

Each frequency ξ evolves independently:

$$e^{ix \cdot \xi} \longrightarrow e^{ix \cdot \xi} \cos(c|\xi|t), \quad e^{ix \cdot \xi} \sin(c|\xi|t).$$

Equivalently, combine exponentials:

$$e^{i(x \cdot \xi \pm c|\xi|t)}.$$

These are **plane wave solutions**.

Thus:

- Solutions are superpositions of plane waves.
- Frequencies travel with speed c .
- Propagation determined by the characteristic cone

$$\tau = \pm c|\xi|$$

The symbol governs the evolution!

Comparison: Wave vs Heat Equation

Wave equation:

$$u_{tt} - c^2 \Delta u = 0$$

Fourier transform gives:

$$\hat{u}_{tt} + c^2 |\xi|^2 \hat{u} = 0.$$

Plane wave solutions:

$$e^{i(x \cdot \xi \pm c|\xi|t)}.$$

Symbol:

$$\tau^2 - c^2 |\xi|^2.$$

Real dispersion relation:

$$\tau = \pm c|\xi|.$$

Oscillatory propagation.

Comparison: Heat Equation

Heat equation:

$$u_t - \kappa \Delta u = 0.$$

Fourier transform gives:

$$\hat{u}_t + \kappa |\xi|^2 \hat{u} = 0.$$

Solution:

$$\hat{u}(\xi, t) = e^{-\kappa |\xi|^2 t} \hat{u}_0(\xi).$$

Plane wave evolution:

$$e^{ix \cdot \xi} \longrightarrow e^{ix \cdot \xi} e^{-\kappa |\xi|^2 t}.$$

Symbol:

$$\tau + \kappa |\xi|^2.$$

Exponential decay of high frequencies.

Symbol and Physics

- Wave equation:
real dispersion relation $\Rightarrow$ propagation along cones.
- Heat equation:
imaginary dispersion $\Rightarrow$ exponential damping.
- Laplace equation:
no time variable $\Rightarrow$ boundary value problem.

The principal symbol determines the qualitative behavior.

Symbol of a Quasilinear PDE

A second-order **quasilinear** PDE has the form

$$\sum_{|\alpha|=2} a_{\alpha}(x, u, \nabla u) D^{\alpha} u + \text{lower order terms} = 0.$$

The **principal symbol** at a solution u is

$$p_2(x, \xi; u) = \sum_{|\alpha|=2} a_{\alpha}(x, u, \nabla u) \xi^{\alpha}.$$

Key point:

- The symbol depends on $(x, u, \nabla u)$.
- Classification (elliptic/hyperbolic/parabolic) may vary with the solution.
- Linearization determines the effective symbol.

Example: Minimal Surface Equation

Minimal surface equation:

$$(1 + u_y^2)u_{xx} - 2u_x u_y u_{xy} + (1 + u_x^2)u_{yy} = 0.$$

Principal symbol:

$$p_2(\xi; u) = (1 + u_y^2)\xi_1^2 - 2u_x u_y \xi_1 \xi_2 + (1 + u_x^2)\xi_2^2.$$

Discriminant:

$$b^2 - ac = (u_x u_y)^2 - (1 + u_y^2)(1 + u_x^2) < 0.$$

Thus the equation is **elliptic** for all smooth u .

Ellipticity is determined by the symbol evaluated at the solution.

Nonlinear Wave Example with Type Change

Consider the quasilinear equation

$$u_{tt} - (1 + u)\Delta u = 0.$$

Freeze at a background solution $u = u_0$.

Linearized principal part:

$$v_{tt} - (1 + u_0)\Delta v.$$

Principal symbol:

$$p_2(\tau, \xi; u_0) = \tau^2 - (1 + u_0)|\xi|^2.$$

Classification depends on u_0 :

- $1 + u_0 > 0 \quad \Rightarrow$ hyperbolic.
- $1 + u_0 = 0 \quad \Rightarrow$ degenerate.
- $1 + u_0 < 0 \quad \Rightarrow$ elliptic in time.

Example: A Nonlinear Wave Equation

Consider the quasilinear equation

$$u_{tt} - c(u)^2 \Delta u = 0,$$

where $c(u) > 0$ is a smooth function.

This is a nonlinear modification of the wave equation: the propagation speed depends on the solution itself.

Principal part:

$$u_{tt} - c(u)^2 \Delta u.$$

Coefficients now depend on u .

Symbol and Hyperbolicity

Freeze the equation at a background solution $u = u_0$.

The linearized principal part is

$$v_{tt} - c(u_0)^2 \Delta v.$$

Principal symbol:

$$p_2(\tau, \xi; u_0) = \tau^2 - c(u_0)^2 |\xi|^2.$$

Characteristic relation:

$$\tau^2 = c(u_0)^2 |\xi|^2.$$

- If $c(u_0)^2 > 0$: hyperbolic.
- If $c(u_0) = 0$ degenerate.